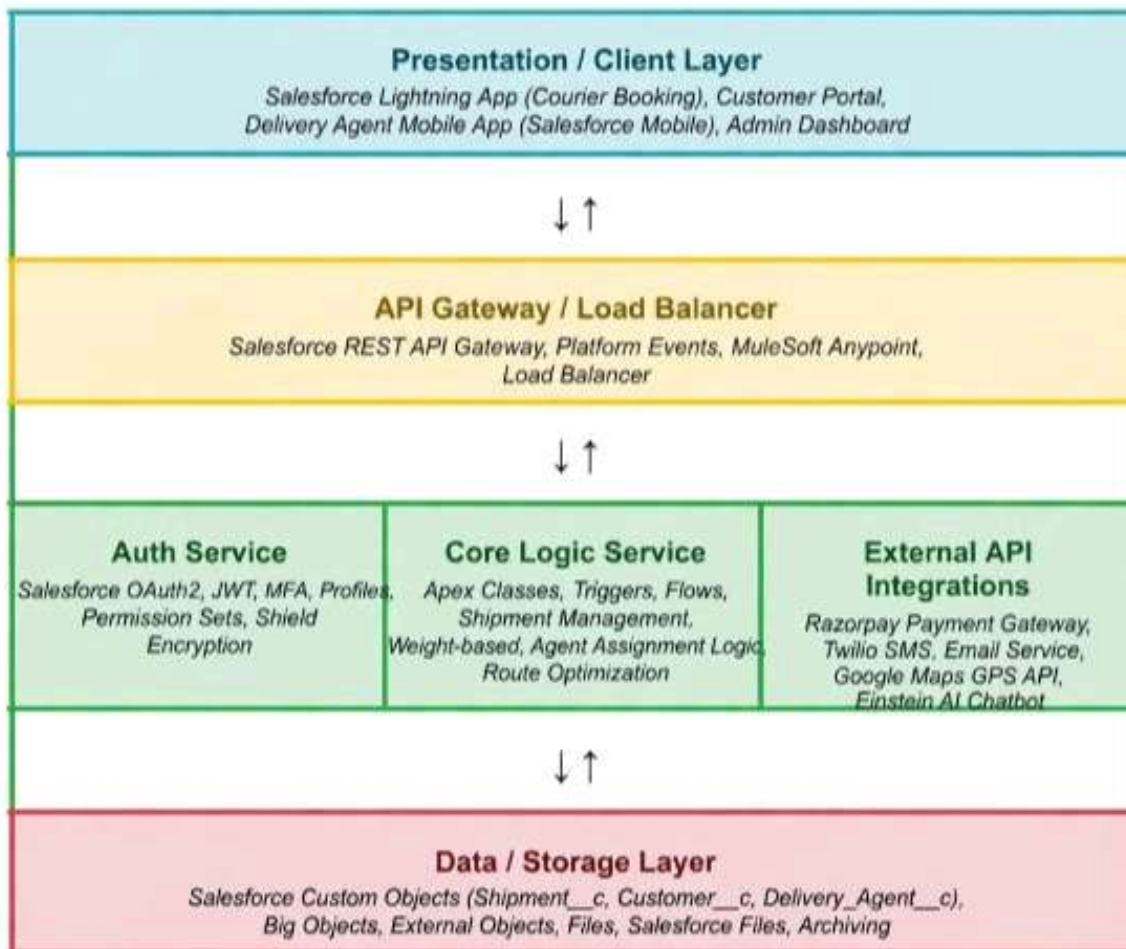


Solution Architecture

Date	29-09-2026
Team ID	SWTID-2026-5359
Project Name	Courier Management System Using Salesforce
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide high-level visual representation of Courier Management System Using Salesforce solution's architecture. Maps out Presentation Layer, Application Layer, and Data Layer for courier operations with booking, tracking, and delivery management.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer - Lightning App, Customer Portal, Agent Mobile App, Admin Dashboard	User interface for courier operations. Customers book shipments & track real-time via portal. Agents update delivery status via Salesforce Mobile App. Admin manages all shipments, customers, agents, reports & dashboards.	Salesforce Lightning Experience, LWC, Salesforce Mobile App, Community Cloud, Customer Portal, Lightning App Builder
API Gateway / Load Balancer	Manages all API requests & load balancing across multiple Salesforce orgs. Handles Platform Events for real-time tracking updates. Ensures scalability to 10,000+ users during peak courier season with load balancers, reports & dashboards.	Salesforce REST API, Platform Events, MuleSoft Anypoint, API Gateway, Load Balancer, CDN
Auth Service	Handles authentication & authorization for courier system. Manages login, Profiles, Permission Sets for Admin, Agent, Customer roles. Ensures security with MFA and Shield encryption for sensitive customer and shipment data.	Salesforce OAuth2, JWT, MFA, Profiles & Permission Sets, Shield Platform Encryption, Health Check, Session Management
Core Logic Service - Shipment Logic, Pricing Engine, Agent Assignment	Main business logic of courier system. Manages shipment creation, weight-based pricing calculation, automated agent assignment, route optimization, notification triggers via Flows, validation rules, and business process automation.	Apex Classes, Triggers, Salesforce Flow, Process Builder, Validation Rules, Einstein AI, Platform Cache, Batch Apex, Queueable Apex
External API Integrations	Integrates third-party services for complete courier operations. Handles payment processing, SMS/Email notifications, GPS navigation for agents, AI chatbot for customer support and real-time shipment tracking.	Razorpay Payment Gateway, Twilio SMS API, SendGrid Email API, Google Maps API, Einstein Bot, Chatbot, Webhooks
Data / Storage Layer - Custom Objects, Big Objects, Files	Stores all courier data securely. Manages Shipment records, customer details, agent information, delivery history, files, invoices, and documents. Handles large data volume with archiving and Big Objects for performance optimization.	Custom Objects: Shipment__c, Customer__c, Delivery_Agent__c, Salesforce Files, Big Objects, External Objects, Data Archiving, Reports & Dashboards